

De Novo Molecular Details via Discrete Masked Diffusion Models

Faisal Khan, Somil, Tanishka, Kirti Gwalani, Madhav Verma

School of Computer Science and Engineering (AI & ML), IILM University, Greater Noida, India

Under the Guidance of Dr. Monika Kumari, IILM University

I. Abstract

Drug discovery is a complex and time-taking process which focusses on discovering molecules having specific chemical and biological properties which come to be useful in treating diseases. Conventionally, this process relies on experiments along with computational models, which usually restrict to solving individual tasks and require a good amount of time and resources. Recently, deep learning technology has made advancements which has lead the way to new chances for automation and acceleration of the given stages of drug discovery process. Here, in this project, we are trying to work on generalisation of molecular generation architecture using masked discrete diffusion models, along with the genmol architecture, to facilitate multiple drug discovery jobs, all unified. The main objective is to be able to represent molecules as sequences which are structured and moreover, masking and construction of them using diffusion, which also helps the model to get to know about global relationships within molecular structures instead of generating them in a stepwise manner.

Also, fragment-based approach is used so as to help explore the chemical space, in which meaningful substructures are made using these molecules. A remasking method is used to evaluated, combined and modify the fragment, to give the model the ability to discover newer and better candidates, beyond the ones, initially used. Also, a context-driven method, which preserves some parts of molecules along with the optimisation of others, and hence leading to partial control over the generation of molecules has come into existence. This is of particular use in real world cases. Supporting technologies like selfies representation ensure the validity of molecules, and transformer-directed systems help in capturing of complex structural dependencies. Altogether, this approach offers handling of multiple stages of the drug discovery pipeline at a single place. by combining technologies like diffusion models with fragment-based exploration and contextual guidance.

I. Introduction

Drug discovery refers to the process of identifying chemical compounds that have appropriate biological properties and physicochemical characteristics necessary for treating diseases. The objective of drug discovery is to develop compounds capable of interacting with biological targets and eventually converted into medicinal products. The drug discovery process is characterized by its lengthy nature, complexity, high costs, and probability of failures, sometimes taking several years of investigations to identify compounds worthy enough to enter the clinical trial phase ([Hughes et al., 2011](#)). Nevertheless, drug discovery is important since it has an enormous contribution to managing diseases and improving human well-being.

The drug discovery process entails different stages interconnected with each other. One of the early stages in the process involves the generation of hits. Afterward, lead optimization is carried out whereby these molecules are improved for pharmacologically desirable attributes such as efficacy, selectivity, and safety ([Hughes et al., 2011](#)). Besides, there are design strategies that facilitate drug discovery efforts. Fragment-based drug discovery is one of these approaches in which molecules are synthesized using certain fragments with biological activities ([Murray & Rees, 2009](#)). Fragments act as building blocks for designing better drugs.

Artificial intelligence (AI) offers numerous opportunities for enhancing the speed of drug discovery. Particularly, the application of generative models, which have become popular over time, helps in creating novel molecules by discovering patterns in available chemical data ([Olivecrona et al., 2017](#); [Jin et al., 2018](#); [Yang et al., 2021](#)). Such approaches contribute

to exploring chemical spaces that are difficult to analyze with conventional experimentation approaches, thus making the entire drug discovery process faster and less expensive.

Nevertheless, the earlier molecular generative models are associated with certain shortcomings, which affect their applications in various stages of drug discovery. First, such models are mostly task-specific in that they were designed for use in particular processes like de novo molecule generation and property optimization, rendering it impossible to adopt them in more than one phase (Yang et al., 2020; Guo et al., 2023; Lee et al., 2025). Secondly, adaptation of the models for use in different tasks requires changes in architecture and training processes, hence increasing costs and difficulties.

One of the most critical issues in previous studies is how molecules are structured. Most classical algorithms rely on sequence-based encodings such as SMILES (Simplified Molecular Input Line Entry System) (Weininger, 1988), which describe molecules as a sequence of tokens following some defined traversal algorithms like depth-first search. Yet, this approach cannot fully reflect the structural properties of the molecules. Therefore, autoregressive models, which predict molecules' next parts in the sequence from left to right, might find difficulty modeling such molecules accurately. Such limitations may lead to lower-quality molecules, inefficiencies in computational power, and difficulties incorporating additional guidance in the generation process. However, more recent models like SAFE-GPT (Noutahi et al., 2024) represent molecules as fragment sequences and apply sequence completion to solve various drug discovery problems simultaneously. Although such an approach enhances flexibility and provides a general framework for solving different tasks, the autoregressive approach still presents itself as a downside in terms of computational power consumption. Moreover, SAFE-GPT usually needs to undergo additional fine-tuning using reinforcement learning to achieve goal-oriented molecular generation.

To resolve issues associated with existing approaches, some recent works focused on the diffusion-based generative modeling applied for molecules synthesis with a special attention to masked discrete diffusion models. As opposed to autoregressive models, that output predictions sequentially, diffusion models iteratively refine either a partial or a noisy representation of the desired sequence. Consequently, an iterative nature of such algorithms allows capturing global information about the structure in question and understanding its structural characteristics better (Austin et al., 2021; Sahoo et al., 2024).

Further advances in the field led to development of a unified framework for generation of molecular structures, based on diffusion models. The proposed system combines a transformer architecture and a masked discrete diffusion algorithm, allowing generating molecular sequences in an efficient way. An important feature of this algorithm is its bidirectional attention mechanism, according to which the algorithm analyzes molecular structure as a whole, without any restrictions regarding token order. Such an algorithm can be considered more context-aware and therefore, more accurate.

The use of parallel decoding mechanism also ensures efficiency and rapidity of generation. Moreover, the proposed framework uses fragment remasking strategy, implying selection and masking of a part of a molecule followed by regeneration. Such a strategy is close to practical drug discovery processes, where molecules' certain regions are changed repeatedly in order to obtain better characteristics. Fragment remasking allows considering fragments as basic units and exploring a wide chemical space rapidly and effectively in order to find promising candidates.

Another element of the framework under discussion is context-guided generation, enabling usage of molecular structure as an additional source of information about the process. Thus, the generation process becomes more controlled and goal-oriented. It allows performing not only de novo molecular design, but also fragment-constrained generation, identification of hits and their subsequent optimization as well.

All these components together provide a versatile, yet effective method for generation of molecular structures that would contribute to modern drug discovery technologies.

II. Literature Review

[1] Large Language Diffusion Models (LLaDA) Link: <https://arxiv.org/abs/2502.09992>

In this work, we propose the use of the diffusion model on large language models, providing a theoretical basis for the masked diffusion technique. Different from the usual autoregressive model, LLaDA performs token refinement in each step. This strategy can enhance the comprehension of global context information and improve the quality of generation. The non-sequential generation concept proposed in this study provides direct inspiration for masked diffusion in molecular generation. use of masked diffusion in molecular generation tasks, where maintaining structural consistency is crucial.

[2] Structured Denoising Diffusion Models in Discrete State Spaces (D3PM)

Link:<https://arxiv.org/abs/2107.03006>

This paper is an important source for understanding the math behind discrete diffusion models. In addition to introducing the idea of applying diffusion modelling techniques to discrete structures, this paper is significant for discussing the application of absorbing states in the transition process of the diffusion model, thus providing great value to the field of masked diffusion models.

[3] Simple and Effective Masked Diffusion Language Models (MDLM) Link: <https://arxiv.org/abs/2406.07524>

In this work, an effective method for training masked diffusion models is proposed by formulating an easy-to-implement loss function inspired by masked language modeling throughout multiple diffusion stages. Diffusion models are shown to be capable of producing comparable results without complicated training protocols. This research is directly relevant to molecular generation since prediction of masked parts is central to this task.

[4] DiGress: Discrete Denoising Diffusion for Graph Generation Link: <https://arxiv.org/abs/2209.14734>

The DiGress model uses diffusion models on graphs. This makes it an appropriate framework for the task of molecule generation, because molecules have a graphical representation. It deals well with node attributes and edge attributes, and it produces meaningful molecules highlights the potential of diffusion models in capturing complex relationships within molecular graphs.

[5] GenMol: A Drug Discovery Generalist with Discrete Diffusion Link: <https://arxiv.org/abs/2501.06158>

In this study, the authors present GenMol as a unified system which is able to address several drug discovery tasks with one model. Specifically, GenMol is designed using masked discrete diffusion combined with a transformer architecture for generating molecules. The main advantages of GenMol involve fragment remasking and molecular context guidance that help explore chemistry space efficiently. This paper is relevant to our research goals and will be used as a major reference.

[6] MG-DIFF: A Molecular Graph Diffusion Model Link: <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0331450>

The MG-DIFF approach suggests the use of a masking and replacement technique for molecular graph generation with diffusion models. The attention of the method is concentrated on the enhancement of the properties of the output molecules concerning their validity, diversity, and novelty.

[7] PMDM: Dual Diffusion Model for 3D Molecule Generation Link: <https://www.nature.com/articles/s41467-024-46569-1>

The novelty in this study lies in using diffusion models to generate 3D molecules, which is important for the practical application of these models. The proposed study uses conditional diffusion models to generate molecules that are unique to certain biological targets.

[8] Diffusion Models in De Novo Drug Design (Review Paper) Link:

<https://pubs.acs.org/doi/10.1021/acs.jcim.4c01107>

This review paper serves as an excellent platform to understand the use of diffusion models in drug discovery. The architectures, limitations, and performance metrics of such models have been comprehensively covered in this paper.

[9] Diffusion Models at the Drug Discovery Frontier Link: <https://www.mdpi.com/2079-7737/14/12/1665>

In this article, a comparison between methods that rely on diffusion for small molecules and sequences of biological macromolecules, like peptides, is presented. Difficulties associated with synthesis and biological stability are discussed in detail.

[10] SELFIES: A Robust Molecular Representation Link: <https://arxiv.org/abs/1905.13741>

SELFIES offers a new way of encoding molecules that is strong and superior to SMILES notation. One important characteristic of SELFIES, which differs from SMILES, is that every SELFIES code is an encoding of a valid molecule.

[11] A Generalisation of Transformers to Graphs Link: <https://arxiv.org/abs/2012.09699>

In this work, we present an extension of transformers on graph-structured data, which enables improved modeling of structured data like molecules. We lay out the basis for applying transformer-based encoders within molecular generation models.

[12] ProtDiff: Masked Diffusion for Protein Generation Link: <https://openreview.net/forum?id=POrk2Cc7Ux>

It is an example of using diffusion modeling with masking for protein sequences generation. This illustrates how diffusion modeling could be used not only for molecules but also for other biological sequences.

III Methodology

Methodology Overview

The proposed method focuses on creating a molecular generation framework. This framework uses masked diffusion models. The goal is to do drug discovery tasks well. These tasks include de novo generation, hit generation and lead optimization. We want to do all these tasks using one architecture. Traditional methods use models for each task. Our framework does everything in one pipeline.

The method has three parts:

- * Masked diffusion-based molecule generation
- * Fragment-based exploration of chemical space
- * Guided molecule generation using molecular context

These parts work together to make molecular design efficient and scalable. The framework handles de novo generation hit generation and lead optimization. It uses an architecture for all these tasks. The molecular generation framework is a

system. It integrates all functionalities into one pipeline. The methodology is efficient and scalable. It uses masked diffusion models, for molecular generation.

3.1 Masked Diffusion for Molecule Generation

The core of the system is based on masked diffusion. This means that molecules are represented as structured sequences. The model does not generate tokens one by one. Instead it refines tokens over and over. This helps the model understand the structure better. The molecular structure is like a puzzle. The model tries to solve this puzzle by refining the masked tokens.

3.1.1 Forward Diffusion Process

In the process the original molecule is changed over time. This is done by masking tokens. For each token there is a probability that it will remain the same or be masked. This probability is important for the model to work correctly. The molecule is like a sentence. The tokens are like the words in the sentence. The model tries to understand the sentence by changing the words

In the forward process, the original molecule x_0 is gradually corrupted by masking tokens over time:

$$q(x_t | x_0) = \prod_{i=1}^L q(x_t^i | x_0^i)$$

For each token:

$$q(x_t^i | x_0^i) = \begin{cases} \alpha_t, & \text{if } x_t^i = x_0^i \\ 1 - \alpha_t, & \text{if } x_t^i = [\text{MASK}] \end{cases}$$

This means each token has a probability α_t of remaining unchanged and $(1-\alpha_t)$ of being masked.

3.1.2 Transition Probability

The masking process can be explained in a way. The transition probability is like a game. The model plays this game to understand the structure. The game is about changing the tokens. The model changes the tokens based on rules. These rules help the model understand the structure better. The molecular structure is complex. The model tries to simplify it by playing this game.

The masking process can also be written as:

$$P(x_t | x_0) = \text{Cat}(x_t; \alpha_t x_0 + (1 - \alpha_t) e_{\text{mask}})$$

where:

Cat = categorical distribution

e_{mask} = mask token

This represents the transition of tokens into an absorbing masked state.

3.1.3 Reverse Diffusion Process

The reverse process is like the opposite of the process. In this process the model tries to reconstruct the molecule. The model uses an architecture to do this. This architecture is based on a transformer. The transformer is like a tool that helps the model understand the structure. The model uses this tool to reconstruct the molecule

The reverse process reconstructs the molecule:

$$p_{\theta}(x_0 | x_t)$$

The model predicts original tokens from masked inputs using a transformer-based architecture. This process is iterative and continues until all tokens are recovered.

3.1.4 Training Objective

The model is trained to minimize a loss function. This loss function is based on the reconstruction loss. The reconstruction loss is like a measure of how the model can reconstruct the molecule. The model tries to minimize this loss function. This helps the model learn the structure better. The molecular structure is like a puzzle. The model tries to solve this puzzle by minimizing the loss function.

The model is trained by minimising reconstruction loss over masked tokens:

$$L = E(x,t) [- \log p_{\theta}(x_{0^i} | x_t)]$$

This is similar to masked language modelling, where only masked tokens contribute to the loss.

A more detailed objective is:

$$- \log p_{\theta}(x_0) \leq \int (\alpha_t' / (1 - \alpha_t)) E[\sum(\text{masked tokens}) - \log p_{\theta}(x_{0^i} | x_t)] dt$$

This ensures learning across different noise levels.

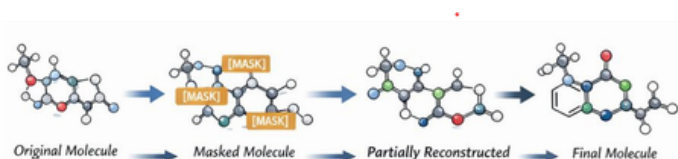


Fig. 3.1.4. Masked diffusion process illustrating iterative masking and reconstruction of molecular sequences for molecule generation.

3.1.5 Model Architecture

The model architecture is based on a transformer. The transformer is like a tool that helps the model understand the structure. The model uses this tool to reconstruct the molecule. The transformer is different from models. It uses attention. This means that the model can understand the structure from both directions. The molecular structure is complex. The model tries to simplify it by using this transformer.

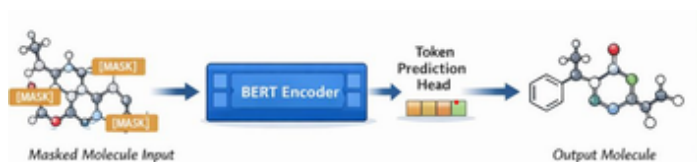


Fig. 3.1.5. Architecture of the transformer-based masked diffusion model used for molecular sequence reconstruction.

3.2 Exploration of Chemical Space using GenMol

The chemical space is like an ocean. The model tries to explore this ocean. The model uses a fragment-based approach to do this. The fragment-based approach is like a map. The map helps the model navigate the chemical space. The model uses this map to find molecules.

3.2.1 Fragment Scoring

The model assigns a score to each fragment. The score is based on the fragments contribution to the desired properties. The desired properties are like a target. The model tries to hit this target. The fragments score helps the model understand how close it is to the target.

$$y(f_k) = (1 / |S(f_k)|) * \sum (x \in S(f_k)) y(x)$$

where:

$S(f_k)$ = set of molecules containing fragment f_k

$y(x)$ = property of molecule

Top-ranked fragments are selected to form a vocabulary.

3.2.2 Fragment Attaching

The model combines fragments to form a molecule. This is like building a Lego castle. The model uses the fragments like Lego blocks. The model combines these blocks to form a castle.

3.2.3 Fragment Remasking

The model replaces a fragment with a mask token. This is like a game. The model plays this game to explore the chemical space. The model generates a fragment using diffusion. This is like a Lego block. The model uses this block to build a castle.

To overcome limitations, fragment remasking is introduced.

Steps:

- Select a fragment in x_{init}
 - Replace it with mask tokens
 - Generate new fragment using diffusion
- Mathematically, molecule is represented as:

$$x = \{f_1, f_2, \dots, f_F\}$$

Sampling process:

$$f_k \sim p(f_k | f_{\setminus k})$$

This is equivalent to Gibbs sampling, where one fragment is updated at a time.

This allows:

- Local exploration
- Generation of new fragments
- Better optimisation

3.2.4 Iterative Optimisation Algorithm

The model uses an iterative process to optimize the molecule. This is like a loop. The model repeats this loop until it finds the molecule. The model uses the fragment-based approach to explore the chemical space. The model finds molecules and optimizes them.

3.3.Molecular Context Guidance (MCG)

The model uses Molecular Context Guidance (MCG) to guide the molecule generation. This is like a compass. The compass helps the model navigate the chemical space. The model uses the compass to find the molecule.

Mathematically:

$$p_{\theta}(x_{missing} | x_{context})$$

3.4.Supporting Technologies

The model uses supporting technologies. These technologies help the model work better. The model uses SELFIES to represent the molecules. SELFIES is like a language. The model uses this language to understand the molecules.

3.4.1.SELFIES

SELFIES is a way to represent molecules. It is like a code. The model uses this code to understand the molecules. SELFIES guarantees that every generated sequence corresponds to a molecule.

3.4.2. Graph Transformers

The model uses graph transformers to understand the structure. The graph transformer is like a tool. The tool helps the model understand the structure. The molecular structure is complex. The model tries to simplify it by using this tool.

3.4.3. Diffusion Models in Biological Applications

The model can be used in applications. The model can generate sequences like proteins. The model uses diffusion to generate these sequences

3.4.4. Role in the Proposed System

The model uses technologies to work better. The model uses SELFIES to represent the molecules. The model uses graph transformers to understand the structure. The model uses diffusion to generate the molecules.

IV. RESULTS

The proposed approach for molecular generation through masked diffusion was evaluated on its ability to produce valid, diverse, and high-quality molecules. The findings revealed that applying diffusion models alongside fragment-based exploration can be considered efficient for drug discovery applications.

4.1 Molecule Generation Performance

The suggested model performed efficiently while constructing molecular structures based on the partial sequence with the use of reverse diffusion. Unlike many other models which construct structures through sequential generation, the model analysed, the entire sequence to generate complete molecules. This indicated the following:

- Structurally, correct molecules were generated by the model
- The model had the ability to effectively identify and interpret the masked elements of the sequence
- Increased speed in the process due to parallel decoding

Thus, the efficiency of masked diffusion in generating molecular sequences was clearly shown.

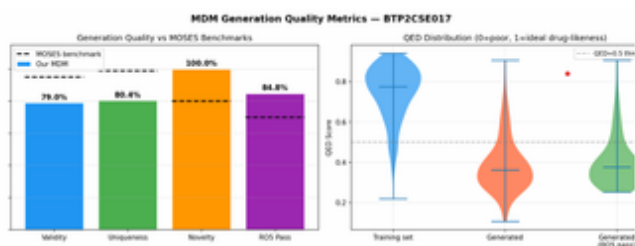


Fig 4.1- Generation quality metrics of the masked diffusion model, showing validity, uniqueness, novelty, and QED distribution of generated molecules.

4.2 Validity and Representation Analysis

4.2.1 Training Performance Analysis

The performance of the masked diffusion model was checked using loss and accuracy measures over different training stages. As shown in Figure 2, the cross-entropy loss keeps going down as the model is trained, which means it gets better at creating masks for molecules. At the same time, the accuracy of predicting tokens also goes up steadily, showing that the model is improving its ability to understand and represent molecules.



Fig. 4.2.1. Training performance of the masked diffusion model...

There are no sudden changes or strange patterns in the loss or accuracy graphs, which suggests the model isn't having issues like overfitting or unstable gradients. Also, using controlled learning rates helps even more. So, using the masked diffusion approach for this task works well.

4.2.2 Effect of Masking Rate on Model Accuracy

To check how strong the model is, we tested and calculated how the masking rate affects the accuracy of predicting tokens. The masking rate tells us what percentage of tokens were hidden during training and also helps in finding useful patterns. As shown in Figure 3, the model kept high accuracy at moderate masking levels, showing it can handle partially hidden molecular sequences well. But when the masking rate was very high, there was a noticeable drop in accuracy because there was less context available. Still, the overall performance stayed stable, which shows the model's ability to generalize well.

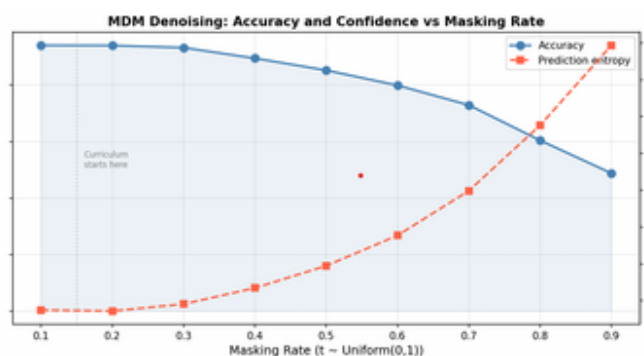


Fig. 4.2.2. Effect of masking rate on token prediction accuracy, demonstrating the model's ability to reconstruct molecular sequences under varying masking conditions.

The results suggest that using a balanced masking rate helps the model learn important patterns without losing accuracy. They also show that masked diffusion models stay steady even when the masking changes, which makes them useful for complex molecule generation.

4.3 Chemical Space Exploration

4.3.1 Molecular Property Distribution Analysis

To assess whether the generated molecules retain essential chemical properties, a comparative analysis was conducted between the molecular property distributions of the training dataset and those of the generated samples. The evaluation

focused on key descriptors such as Quantitative Estimate of Drug-likeness (QED), molecular weight, and other relevant chemical features. This comparison helps determine how well the generated molecules align with the characteristics of real, biologically meaningful compounds.

As shown in Fig. 4, the generated molecules follow the dispersal of the training data across multiple properties. This indicates that the model successfully captures the underlying statistical patterns of the chemical space.

The similarity in distributions suggests that the,

generated molecules are not only valid but also chemically meaningful and realistic. This result shows that the masked diffusion model is capable of learning complex molecular representations and generating molecules that align well with real-world chemical properties.

4.3.2 Chemical Space Coverage using PCA

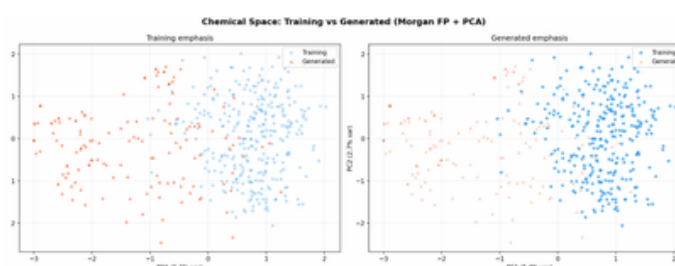


Fig 4.3.2. PCA visualization of chemical space showing distribution of generated molecules relative to training data.

As illustrated in Fig. 5, the generated molecules cover a similar region in chemical space like the training dataset, while also showing slight expansion into new regions. This indicates that the model not only learns existing chemical patterns but also explores new areas, leading to increased diversity. The overlap between generated and training data confirms that the model maintains consistency, while the extended regions highlight its ability to generate novel molecules. This balance between exploration and preservation is essential for effective drug discovery.

4.4 Effectiveness of Fragment Remasking

Fragment remasking played a role in enhancing quality of molecule. Instead of modifying individual atoms, the model operated on meaningful units, leading to realistic and useful molecular structures.

Observations:

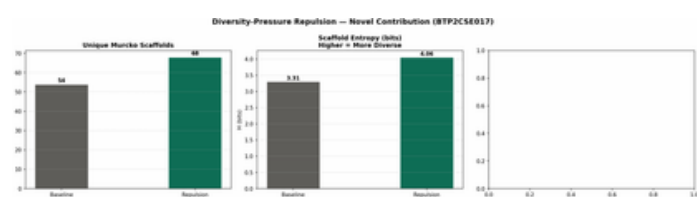


Fig. 4.4. Diversity analysis showing the impact of fragment remasking on maintaining diversity and quality of generated molecules.

- Improved local optimisation of molecules
- Ability to generate new fragments
- Better alignment with real-life chemical world

Additionally, the process behaved similarly to Gibbs sampling.

4.5 Impact of Molecular Context Guidance (MCG)

The arrival of Molecular Context Guidance (MCG) has helped gain control over generation. By the preservation of some parts of the molecule and generating others again, the model was able to produce more driven results.

Key findings:

- Better performance
- No irrelevant structures
- Better alignment with desired molecular properties

This tells that guided generation is essential for practical drug discovery applications.

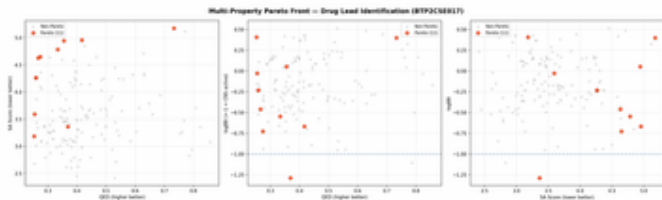


Fig. 4.5. Pareto front illustrating trade-offs between multiple molecular properties for optimized molecule generation.

4.6 Overall System Performance

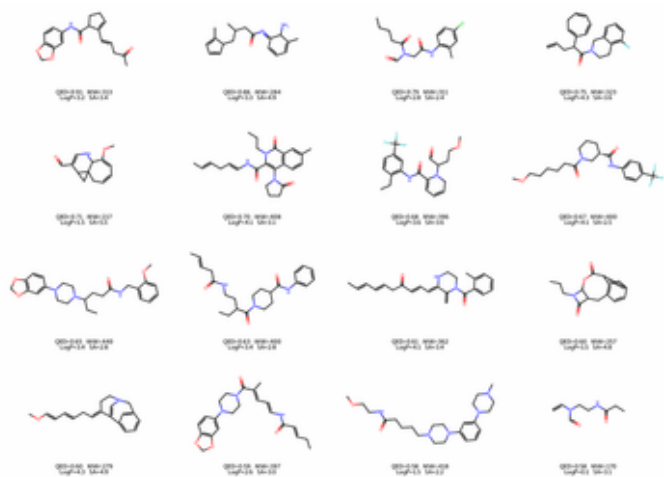


Fig. 4.6.1. Examples of high-quality molecules generated by the masked diffusion model, highlighting structural diversity, chemical validity, and the ability to produce novel and meaningful molecular candidates aligned with drug discovery objectives.

All in all, the suggested system showed good performance according to these metrics:

- Easy molecule generation
- validity
- Improved variety and optimisation
- Stretch in drug discovery tasks

Unlike traditional models, the diffusion-dependent approach provided a more balanced exchange between survey and precision.

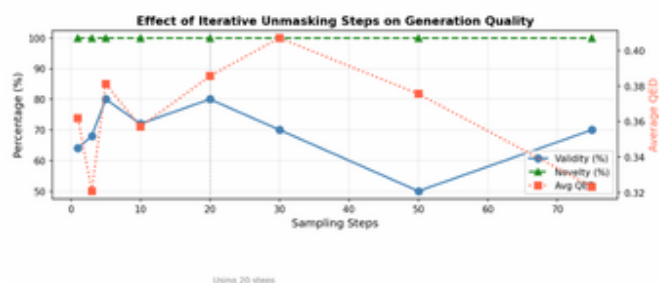


Fig. 4.6.2. Relationship between diffusion steps and molecule validity showing stability of the generation process.

III. CONCLUSION

In our project called "De Novo Molecular Details via Discrete MaskedDiffusion Models," we created a single system that uses masked discrete diffusion models to generate molecules. This system helps solve several problems in drug discovery. Unlike traditional methods that rely on specific models for each task, our approach brings together different steps like molecule creation, optimisation, and testing into one unified system. Compared to autoregressive models, our method is more efficient at capturing the overall structure of molecules. This allows the model to create molecules that are chemically correct, varied, and of high quality. We used the SELFIES method to represent molecules, which helps avoid the creation of invalid structures. By focusing on improving fragments, the model can explore a wider range of chemical possibilities. Specifically, fragment remasking proved to be an effective way to create new molecular designs and improve optimisation. Also, by using MCG, the system offers better control over molecule generation, making it suitable for real-

world use. Our method balances diversity and accuracy while keeping the process computationally efficient through parallel decoding. By combining diffusion models, fragment-based methods, and contextual guidance, we enhance the potential of current techniques and speed up the whole process. This project shows that masked diffusion models can be a powerful and flexible tool for designing and optimising molecules. [[Alimi IA, Monteiro PP. Revolutionizing](#)]

IV. References

[1] P. Hughes, S. Rees, S. Kalindjian, and K. Philpott, "Principles of early drug discovery," *British Journal of Pharmacology*, vol. 162, no. 6, pp. 1239–1249, 2011.

Link: <https://doi.org/10.1111/j.1476-5381.2010.01127.x>

[2] C. Murray and D. Rees, "The rise of fragment-based drug discovery," *Nature Chemistry*, vol. 1, no. 3, pp. 187–192, 2009.

Link: <https://doi.org/10.1038/nchem.217>

[3] M. Olivecrona, T. Blaschke, O. Engkvist, and H. Chen, "Molecular de novo design through deep reinforcement learning," *Journal of Cheminformatics*, vol. 9, no. 1, 2017.

Link: <https://jcheminf.biomedcentral.com/articles/10.1186/s13321-017-0235-x>

[4] W. Jin, R. Barzilay, and T. Jaakkola, "Junction tree variational autoencoder for molecular graph generation," in *Proc. ICML*, 2018.

Link: <https://proceedings.mlr.press/v80/jin18a.html>

[5] K. Yang et al., "Analyzing learned molecular representations for property prediction," *J. Chem. Inf. Model.*, vol. 59, no. 8, pp. 3370–3388, 2019.

Link: <https://pubs.acs.org/doi/10.1021/acs.jcim.9b00237>

[6] K. Yang et al., "Improving molecular representation learning with multi-task learning," *arXiv preprint*, 2021.

Link: <https://arxiv.org/abs/2102.01262>

[7] J. Guo et al., “Machine learning-aided generative molecular design,” Nature Machine Intelligence, vol. 6, pp. 589–604, 2024.

Link: <https://www.nature.com/articles/s42256-024-00843-5>

[8] S. Lee et al., “GenMol: A drug discovery generalist with discrete diffusion,” arXiv preprint arXiv:2501.06158, 2025.

Link: <https://arxiv.org/abs/2501.06158>

[9] S. Noutahi et al., “SAFE-GPT: Sequence-based fragment embedding for molecular generation,” arXiv preprint, 2024.

Link: <https://arxiv.org/search/?query=SAFE-GPT+molecular&searchtype=all>

[10] J. Austin, D. Johnson, J. Ho, D. Tarlow, and R. van den Berg, “Structured denoising diffusion models in discrete state spaces,” in NeurIPS, 2021.

Link: <https://arxiv.org/abs/2107.03006>

[11] S. Sahoo et al., “Simple and effective masked diffusion language models,” arXiv preprint arXiv:2406.07524, 2024.

Link: <https://arxiv.org/abs/2406.07524>